



2524 - Trace gases in Titan's Atmosphere with JWST MIRI

Cycle: 1, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>	<i>E-Mail</i>
Dr. Conor Nixon (PI)	NASA Goddard Space Flight Center	conor.a.nixon@nasa.gov
Prof. Patrick Irwin (CoI) (ESA Member)	University of Oxford	patrick.irwin@physics.ox.ac.uk
Nick Teanby (CoI) (ESA Member)	University of Bristol	n.teanby@bristol.ac.uk
Dr. Keeyoon Sung (CoI)	Jet Propulsion Laboratory	keeyoon.sung@jpl.nasa.gov

OBSERVATIONS

ABSTRACT

In this JWST data archive project, we propose to model mid-infrared MIRI MRS spectra of Titan acquired under GTO project #1251 (“Titan Climate, Composition and Clouds”) with the goal of detecting new chemical species in Titan’s atmosphere. In particular, we will focus on the 5-7 micron spectral range, which has not been previously seen by either of Cassini’s infrared spectrometers or other observatories. This spectral range contains molecular bands of many organic molecules previously unseen in Titan’s atmosphere, which may be detected for the first time by virtue of JWST’s very high sensitivity. Such new detections would greatly contribute to our knowledge of Titan’s organic chemistry, and to the wider field of astrobiology.

OBSERVING DESCRIPTION

This proposal makes use of public-archive observations of Titan acquired under GTO program #1251. For MIRI, these are MRS spectra with 555s integration times. Other details of the planned observations are in the public domain.